Module 4: Forest Management | DAY 1

2026-11-09

Introduction

Members of the K'avi tribe have managed a forest on their lands for decades. Recently, tribal forestry managers have started collecting data from a few different sites in the forest. They would like to set up a location for managing and harvesting Douglas firs for lumber. It is your job to analyze the data and determine which of the proposed sites would be the best for a harvest.

The K'avi want to choose a site that is the most productive, or in other words, has the greatest average board feet value for a given time period. Board feet are a unit of measurement that expresses the volume of logs and standing timber.

They have also found throughout the time that they have been caring for the forest, that there are many factors that affect the quality of a given site. Components such as altitude, aspect, soil, and rainfall, as well as the combination of these factors have an effect on the productivity of a site.

1. Altitude affects productivity by shortening the length of the growing season and lowering the mean temperature. At high altitude, growth begins late and stops early, therefore making them less productive.
2. Aspect also affects productivity, as sites that face north, northeast, and east are more fertile (likely due to the sun's rays having less of an effect on the soil's dampness). Aspects that faced south to west had variable affects. Also, sites with south, southwest, and south aspects were less productive.
3. Soil must have the necessary substances to meet the conditions needed to grow. The most productive sites have deep, well-drained sandy loam soils, with the second-most being clay soils, and the least productive being gravel soils.
4. Rainfall, especially when it is abundant, greatly increases the tree growth at a site. The most productive sites have an annual precipitation of over 60 inches with less productive sites having an annual precipitation of less than 30 inches.

Below are short descriptions of the sites the K'avi managers have been collecting data on:

1. Site 1: At an altitude of 2,000 feet, an aspect that faces south to west, and annual precipitation of less than 40 inches
2. Site 2: Faces northeast, has well-drained, sandy loam soil and an altitude of 1,500 feet.
3. Site 3: Rainfall of less than 20 inches annually, a southern aspect, and gravel soil
4. Site 4: Sandy loam soil, an altitude of 1,600 feet and 65 inches of rainfall annually.

Overview of Module 4

- Exploring the data
- Summary statistics
- Choosing a site
- Combining the data
- Calculating statistics from the combined data
- Creating a table
- Graphing
- Recap

Learning Objectives for Module 4:

1. Evaluate both numeric trends and descriptive notes (quantitative and qualitative data) to form a holistic view of forest health
2. Learn to design professional data visualizations with labels, legends, and customized colors to make ready to present to diverse audiences
3. Apply data analysis and visualization to deliver clear, accessible recommendations to forest managers and community
4. Discover where to find helpful online resources to continue exploring data and creating data visualizations independently in the future

Lesson 4.1: Exploring data with ggplot

This lesson will review how we can use ggplot to explore a data frame with continuous variables and one column of a categorical variable.

First, set up the environment.

```
#
rm(list = ls())

#
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

Read in data

Below is the data that the forestry managers have been collecting from the different site:

```
site_1 <- read.csv("site_1.csv")
site_2 <- read.csv("site_2.csv")
site_3 <- read.csv("site_3.csv")
site_4 <- read.csv("site_4.csv")
```

Exploring the Data

Now that you've loaded in the data from the different sites,insert a code chunk below to explore what the variables are in the data:

```
#Use this chunk to look at the data above


#Could also use dim, glimpse, view....
```

Class Questions:

- What are the dimensions of the data from each site?
- What variables does the data you just loaded in contain?
- What do these variables mean in the context of forestry management?
- How could this data be used to determine which site is the best choice for forestry?

Summary statistics

Now lets start to explore the differences between the sites with summary statistics.

Class Question:

- What statistics would help you explore the difference in productivity between the different sites?

You can use the following chunk to perform whichever summary statistics you would find helpful.

```
# Use this chunk to do any summary statistics you feel would be helpful.
```

Class Questions:

- Based on these summary statistics, which site seems like the best site so far?
- Is there any other information you would like to consider about these sites that was not provided in this data set?

Combining the sites

Now that you've loaded in and explored that data from each of the sites, it will be useful to combine it in one large data set. We will use `mutate()`, similar to the previous module, to create a new column to identify each site.

Then we will combine the rows using `rbind()`. This function combines all the data vertically, but in order to do so, all columns must have the same name.

```
# create new column site_ID
site_1_new <- site_1 %>%
  mutate(site_ID = 1)
site_2_new <- site_2 %>%
  mutate(site_ID = 2)
site_3_new <- site_3 %>%
  mutate(site_ID = 3)
site_4_new <- site_4 %>%
  mutate(site_ID = 4)

# combine rows with rbind
forest<- rbind(site_1_new, site_2_new, site_3_new, site_4_new)

# preview it
head(forest)
```

```
##   age board_feet tree_density site_ID
## 1  20      0.000            0       1
## 2  30   6041.434          165       1
## 3  40  17490.338          278       1
## 4  50  31906.178          322       1
## 5  60  44130.086          308       1
## 6  70  56994.003          279       1
```

Calculate site averages

Now calculate the mean board feet at each site from the new combined data. Similar to previous modules, use `group_by()` and `summarize()`

```
#Use this chunk to calculate the mean board_feet from the combined data
forest %>%
  group_by(site_ID) %>%
  summarize(mean_board_feet = mean(board_feet))
```

```
## # A tibble: 4 × 2
##   site_ID mean_board_feet
##     <dbl>           <dbl>
## 1       1          65852.
## 2       2         120153.
## 3       3          29141.
## 4       4         107197.
```

Again using `group_by()` and `summarize()`, create a table/tibble with the `median()` age of the trees at each site.

```
#Use this chunk to calculate the median age of trees from the combined data
```

Exploring data with plots

In order to compare the different sites, we want to use the variable `site_ID` in our plots. Even though we know that it is a discrete value we are using as an ID, R recognizes this variable as a double which will cause issues when trying to plot it as a category. In order to fix this, we are going to save `site_ID` as a factor. Factors in R are categorical variables that can be stored in levels.

```
forest$site_ID <- as.factor(forest$site_ID)
```

Boxplot

We are using two columns to plot a boxplot - `site_ID` is a discrete category, and the `board_feet` is a continuous variable. The boxplots will show the distribution of values and statistics for `board_feet` at each site.

Create a boxplot with the `forest` dataframe showing `site_ID` on the x-axis, and `board_feet` on the y-axis.

Scatter plot

Our data frame also contains an `age` column, which represents years, and could be treated as continuous or discrete. In this case, to create a time series, we can treat it as continous, and use it in a scatter plot. Scatter plots typically show the relationship between two continuous variables.

Create a scatter plot with the `forest` dataframe of `board_feet` v. `age` , with `site_ID` as color or shape, using `geom_point()`

Lesson 4.1 Recap:

You have now explored the data in many different ways. Based on the information from the summary statistics, graph, and descriptions of the sites, which site would you recommend as the best for a lumber harvest and why?